PADMAPRIYA V

ASSINGNMENT 1 -GEN AI – VL [28.05.2026]

1.What is the output of the following, and why?

s "Python"

print(s[0], s[-1], s[6])

Output

```
>>> s="python"
>>> print(s[0],s[-1],s[6])
    Traceback (most recent call last):
      File "<pyshell#1>", line 1, in <module>
        print(s[0],s[-1],s[6])
    IndexError: string index out of range
```

It shows error because string index is out of range

2.What is the difference between positive and negative indexing?

Given s = "Hello", what does s[-2] return?

Output

```
>>>
>>> s="hello"
>>> print(s[-2])
    l
```

3.What happens when you access an index that is out of range in a string?

How is this different from slicing out of range?

S=”code”

print(s[10])

print(s[1:10])

Output

```
>>>
>>> s="code"
>>> print(s[10])
    Traceback (most recent call last):
      File "<pyshell#7>", line 1, in <module>
        print(s[10])
    IndexError: string index out of range
>>> print(s[1:10])
    ode
```

It’s shows error for the first input because the string is out of range.

4.How do you find the index of the first and last occurrence of a character in a string using indexing concepts?

s = "mississippi"

Find index of first 's' and last 's'

Output

```
>>>
>>> s="mississippi"
>>> print(s.find("s"))
    2
>>> print(s.rfind("s"))
    6
```

5.What is the output of the following, and explain how negative indexing resolves?

s = "abcdef"

print(s[-1], s[-3], s[-6])

Output

```
>>>
>>> s="abcdef"
>>> print(s[-1],s[-3],s[-6])
    f d a
```

6.What is the syntax for string slicing in Python? Explain each part of s[start:stop:step].

Start : The number from which the sequence begins.

Default value is 0.

Stop: The number at which the sequence ends.

The stop value is not included in the output.

Step: The amount by which the value increases or decreases each time.

Default value is 1.

7. What is the output of the following slices, and why?

s = "HelloWorld"

print(s[0:5])

print(s[S:])

print(s[:5])

print(s[:])

Output

```
>>> s="hello world"
>>> print(s[0:5])
hello
>>> print(s[5:])
 world
>>> print(s[:5])
hello
>>> print(s[:])
hello world
```

8.What happens when start is greater than stop in a slice with a positive step? What is the output?

s = "Python"

print(s[4:1])

Output

BLANK

Because Starting value always should be less then ending value

9.How do you extract the last 3 characters of a string using slicing, without knowing its length?

Output

```
>>>
>>> s="priya"
>>> print(s[2:5])
iya
```

10. What is the difference between these two slices? What does each return?

s = "abcdefgh"

print(s[::1])

print(s[::2])

print(s[1::2])

output

```
>>>
>>> s="abcdefgh"
>>> print(s[::1])
abcdefgh
>>> print(s[::2])
aceg
>>> print(s[1::2])
bdfh
```

11.How do you reverse a string using slicing? What is the step value that achieves this?

s = "Python"

Reverse it using slicing

Output

```
>>> s="python"
>>> print(s[::-1])
nohtyp
```

12.What is the output of the following, and explain how the negative step works?

s = "abcdefgh"

print(s[::-1])

print(s[::-2])

print(s[6:1:-1])

Output

```
>>> s="abcdefgh"
>>> print(s[::-1])
hgfedcba
>>> print(s[::-2])
hfdb
>>> print(s[6:1:-1])
gfedc
```

13.How do you extract every 3rd character from a string starting from index 1?

s = "a1b2c3d4e5"

Expected output: "12345"

Output

```
>>>
>>> s="a1b2c3d4e5"
>>> print(s[1:10:2])
12345
```

14.What is the output of the following negative-step slices?

Explain start and stop behavior when step is negative.

s = "Python"

print(s[S:0:-1])

print(s[S::-1])

print(s[:0:-1])

Output

```
>>> s="python"
>>> print(s[5:0:-1])
nohty
>>> print(s[5::-1])
nohtyp
>>> print(s[:0:-1])
nohty
>>>
```

15.How do slices handle start and stop values that exceed string boundaries with a negative step?

s = "hello"

print(s[100::-1])

print(s[::-100])

Output

```
>>> s="hello"
>>> print(s[100::-1])
olleh
>>> print(s[::-100])
o
```

16.How do you extract the middle portion of a string dynamically, regardless of its length?

s = "abcdefghij"   # 10 chars

Extract characters from index 2 to second-last (exclusive)

Output

```
>>> s="abcdefghij"
>>> print(s[2:8])
cdefgh
>>>
>>> print(s[2:8])
cdefgh
>>>
```

17.How do you extract the middle portion of a string dynamically, regardless of its length?

s = "racecar"

Check palindrome using slicing

Output

```
>>> s="racecar"
>>> print(s[::-1])
racecar
```

Palindrome means it's a sequence of characters that remains the same when read from left to right and from right to left.

18.What is the output of the following, and why does Python not raise an error?

s = "Hello"

print(s[1:100])

print(s[-100:3])

print(s[100:200)

Output

```
>>>
>>> s="hello"
>>> print(s[1:100])
ello
>>> print(s[-100:3])
hel
>>> print(s[100:200])
```

For the last input the output shows blank because there is no characters given.

19.How do you replace a portion of a string using slicing (since strings are immutable)?

s = "Hello World"

Replace "World" with "Python" using slicing and concatenation

Output

```
>>>
>>> s="hello world"
>>> print(s[:6]+"python")
hello python
>>>
```

20.How does Python's slice() object work, and how can you use it as a reusable slicer?

s = "abcdefghij"

my_slice = slice(1, 8, 2)

print(s[my_slice])

What is the output, and how can this be reused on

other strings?

Output

```
>>>
>>> s="abcdefghij"
>>> my_slice=slice(1,8,2)
>>> print(s[my_slice])
bdfh
>>>
```